

DATA DETECTIVE MISSION

Exploratory Data Analysis Report — UCI Online Retail Dataset

1. Activity Information

Item	Details
Activity Type	Team-based Investigation / EDA Lab
Team Size	3–4 students
Duration	2 Hours
Tools Used	Python, Google Colab, Pandas, NumPy, Matplotlib, Seaborn
Dataset	UCI Online Retail
Dataset Source	UCI Machine Learning Repository

2. Introduction

The **Data Detective Mission** was an Exploratory Data Analysis (EDA) activity based on the **UCI Online Retail dataset**. The dataset contains real transaction records from an online retail business based in the UK.

The main aim of this activity was to understand the data before making any conclusions from it. We looked at the structure of the dataset, checked for missing and duplicate values, identified unusual transactions, and then used different visualizations to find useful patterns.

As junior data analysts, our focus was not just on finding numbers but also on understanding what those numbers could tell us about the business.

3. Real-World Scenario

For this activity, we were given historical sales data from an online retailer. The company wanted us to investigate the data and answer questions such as:

- Which products generate the most revenue?
- How does revenue change from month to month?
- Which countries contribute the most to sales?
- Are there missing or inconsistent records?
- Are there unusual transactions or returns?
- What useful business decisions can be made from the data?

Our task was to **inspect, clean, analyze, and visualize the data** and then use our findings to give a practical business recommendation.

4. Learning Objectives

Through this activity, we aimed to:

1. Understand the structure and important variables in a real-world dataset.
 2. Find missing values, duplicate records, and unusual observations.
 3. Use descriptive statistics to understand the data better.
 4. Create suitable visualizations to identify patterns.
 5. Convert the results of our analysis into meaningful business insights.
-

5. Dataset Description

The **UCI Online Retail dataset contains 541,909 transaction records**.

It includes information about invoices, products, quantities, prices, customers, dates, and countries.

Variable	Description
InvoiceNo	Unique invoice or transaction number
StockCode	Unique product identification code

Description	Name or description of the product
Quantity	Number of items purchased
InvoiceDate	Date and time of the transaction
UnitPrice	Price of one item in GBP
CustomerID	Unique customer identification number
Country	Country of the customer

During preprocessing, we also created a new variable called **Revenue**.

$$\text{Revenue} = \text{Quantity} \times \text{UnitPrice}$$

This helped us understand how much revenue was generated by individual transactions and products.

6. Data Loading and Initial Inspection

We loaded the dataset into **Google Colab** using Python and Pandas.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_excel('/content/Online Retail.xlsx')
```

```
print("Dataset Shape:", df.shape)
display(df.head())
```

Initial Observations

After loading the dataset, we found that it contained:

- **541,909 rows**
- **8 original columns**
- Transaction-level information
- Numerical, categorical, and date/time variables

We also looked at the first few rows to get a basic idea of how the transactions were recorded.

7. Data Dictionary

Before starting the analysis, we prepared a simple data dictionary so that we could clearly understand what each column represented.

Variable	Data Type	Purpose
InvoiceNo	Categorical	Identifies an invoice
StockCode	Categorical	Identifies a product
Description	Categorical	Gives the product name
Quantity	Numerical	Shows the number of products purchased
InvoiceDate	DateTime	Shows when the transaction took place
UnitPrice	Numerical	Shows the price of one product
CustomerID	Categorical	Identifies the customer
Country	Categorical	Shows the customer's country

8. Data Quality Investigation

Before analyzing the sales patterns, we first checked whether the dataset itself had any problems.

We specifically looked for:

- Missing values
- Duplicate records
- Incorrect data types
- Negative quantities
- Invalid prices
- Cancelled transactions
- Extremely large or unusual values

We used the following commands:

```
print("DATA TYPES")
display(df.dtypes)

print("\nMISSING VALUES")
display(df.isnull().sum())

print("\nDUPLICATE ROWS:", df.duplicated().sum())

print("\nSTATISTICAL SUMMARY")
display(df.describe(include="all").T)
```

This step was important because analyzing unclean data could lead to incorrect conclusions.

8.1 Missing Values

We found missing values mainly in:

- CustomerID
- Description

The missing CustomerID values mean that some transactions cannot be linked to a particular customer.

Impact

Because of this, customer-level analysis needs to be done carefully. We cannot assume that every transaction belongs to a known customer.

9. Duplicate Records

We checked for duplicate rows using:

```
df.duplicated().sum()
```

Duplicate records can affect the results by making it appear that:

- More products were sold than actually were
- More revenue was generated
- More transactions took place

They can also affect statistical calculations.

Therefore, duplicate records were removed during the cleaning process.

10. Unusual Transactions

While exploring the dataset, we found transactions with **negative quantities**.

In a retail dataset, a negative quantity usually indicates that products were returned or that a transaction was cancelled rather than representing a normal purchase.

We also noticed that invoice numbers beginning with **C** represented cancellation transactions.

Therefore, these transactions were treated separately from normal sales during preprocessing.

11. Data Preprocessing

Before performing the main analysis, we cleaned the dataset.

The following steps were carried out:

1. Converted **InvoiceDate** into the correct DateTime format.
2. Removed duplicate records.
3. Identified cancelled invoices.
4. Removed transactions with non-positive quantities from the normal-sales analysis.
5. Removed transactions with non-positive unit prices.
6. Created the **Revenue** column.
7. Created additional time-related columns such as month and day.

Preprocessing Code

```
clean = df.copy()
```

```
clean["InvoiceDate"] = pd.to_datetime(clean["InvoiceDate"])
```

```
clean = clean.drop_duplicates()
```

```
clean = clean[  
    ~clean["InvoiceNo"].astype(str).str.startswith("C")
```

```
]
```

```
clean = clean[  
    (clean["Quantity"] > 0) &  
    (clean["UnitPrice"] > 0)  
]
```

```
clean["Revenue"] = clean["Quantity"] * clean["UnitPrice"]
```

```
clean["Month"] = clean["InvoiceDate"].dt.to_period("M").astype(str)
```

```
clean["Day"] = clean["InvoiceDate"].dt.day_name()
```

After this step, the dataset was in a much better condition for analyzing normal sales.

12. Descriptive Statistics

We used descriptive statistics to get a better understanding of the numerical data.

The main measures we looked at were:

- Count
- Mean
- Standard deviation
- Minimum
- Maximum
- Quartiles

The results showed that quantities and prices were not evenly distributed. Some transactions had much larger quantities or values than most other transactions.

These unusual observations can be considered **outliers**, so they need to be kept in mind when interpreting averages.

13. Visualization 1 — Monthly Revenue Trend

Objective

The purpose of this visualization was to see how the company's revenue changed over time.

```
monthly_sales = clean.groupby("Month")["Revenue"].sum()
```

```
plt.figure(figsize=(14,5))
monthly_sales.plot(marker="o")
plt.title("Monthly Revenue Trend")
plt.xlabel("Month")
plt.ylabel("Revenue (£)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

Finding

The graph shows that revenue was not the same every month. Some months had noticeably higher sales activity than others.

Business Insight

The company can use these historical patterns to plan:

- Inventory
- Promotions
- Staffing
- Product availability

This can help the business prepare better for periods of higher demand.

14. Visualization 2 — Top 10 Products by Revenue

Objective

The purpose of this analysis was to find the products that contributed the most revenue.

```
top_products = (
    clean.groupby("Description")["Revenue"]
    .sum()
    .sort_values(ascending=False)
    .head(10)
```

```
        .sort_values()
    )

plt.figure(figsize=(10,6))
top_products.plot(kind="barh")
plt.title("Top 10 Products by Revenue")
plt.xlabel("Revenue (£)")
plt.ylabel("Product")
plt.tight_layout()
plt.show()
```

Finding

The analysis showed that revenue was concentrated among a smaller group of products.

Business Insight

These high-revenue products deserve more attention when it comes to:

- Maintaining stock
- Reordering products
- Promotions
- Forecasting future demand

Keeping popular products available can help prevent missed sales opportunities.

15. Visualization 3 — Top 10 Countries by Revenue

Objective

We used this visualization to understand which countries contributed the most to the company's revenue.

```
country_sales = (
    clean.groupby("Country")["Revenue"]
        .sum()
        .sort_values(ascending=False)
        .head(10)
        .sort_values()
    )
```

```
plt.figure(figsize=(10,6))
country_sales.plot(kind="barh")
plt.title("Top 10 Countries by Revenue")
plt.xlabel("Revenue (£)")
plt.ylabel("Country")
plt.tight_layout()
plt.show()
```

Finding

The **United Kingdom is the dominant market** in the dataset, while several other countries also contribute to international sales.

Business Insight

Looking at sales country-wise allows the company to compare its domestic and international markets and identify areas that may need more attention.

16. Visualization 4 — Quantity vs Unit Price

Objective

The purpose of this visualization was to see how quantity and unit price are distributed and to identify unusual transactions.

```
sample = clean.sample(
    min(10000, len(clean)),
    random_state=42
)
```

```
plt.figure(figsize=(9,6))
```

```
sns.scatterplot(
    data=sample,
    x="UnitPrice",
    y="Quantity",
    alpha=0.4
)
```

```
plt.title("Quantity vs Unit Price")
plt.xlabel("Unit Price (£)")
plt.ylabel("Quantity")
```

```
plt.tight_layout()
plt.show()
```

Finding

Most transactions are grouped within relatively smaller quantity and price ranges. However, there are some transactions with unusually high quantities.

Business Insight

These unusual transactions could be worth investigating because they might represent:

- Bulk purchases
 - Special orders
 - Data-entry errors
 - Exceptionally high customer demand
-

17. Pattern Identification

After completing the main visualizations, we carried out some additional analysis to understand the sales patterns in more detail.

Top Products by Quantity

```
top_quantity = (
    clean.groupby("Description")["Quantity"]
        .sum()
        .sort_values(ascending=False)
        .head(10)
)
```

```
display(top_quantity)
```

This helped us identify the products that were sold in the highest quantities.

Highest-Value Customers

```
top_customers = (  
    clean.groupby("CustomerID")["Revenue"]  
        .sum()  
        .sort_values(ascending=False)  
        .head(10)  
)  
  
display(top_customers)
```

This analysis helped identify customers who contributed significantly to the company's revenue.

However, because some **CustomerID** values are missing, the results should be interpreted with that limitation in mind.

Average Transaction Value

```
invoice_value = clean.groupby("InvoiceNo")["Revenue"].sum()  
  
print("Average transaction value:",  
      round(invoice_value.mean(), 2))  
  
print("Median transaction value:",  
      round(invoice_value.median(), 2))
```

We compared the average and median transaction values to understand whether unusually large transactions were having a noticeable effect on the average.

18. Key Data-Quality Findings

Issue	What We Found	Effect on Analysis
Missing CustomerID	Many transactions do not have customer information	Makes customer-level analysis difficult
Missing Description	Some products have no description	Makes product analysis less clear

Duplicate records	Duplicate rows are present	Can increase calculated sales and transaction counts
Negative Quantity	Returns and cancellations are present	Should not be treated as normal sales
Cancelled invoices	Some invoice numbers begin with C	Need to be handled separately
Extreme values	Some quantities and prices are unusually large	Can affect averages and statistics

19. Key Business Findings

After completing the analysis, we identified several important patterns.

1. Sales change over time

Sales activity is not the same throughout the year. Some months show stronger activity than others.

2. Some products contribute much more revenue

A smaller group of products contributes a significant portion of the overall revenue.

3. The UK is the main market

The dataset shows that the UK accounts for a large share of the sales.

4. Returns and cancellations matter

Negative quantities and cancelled invoices need to be separated from normal sales to avoid misleading results.

5. Customer information is incomplete

Since many transactions do not have a **CustomerID**, complete customer segmentation is not possible using this dataset alone.

6. Outliers need attention

Some transactions have unusually large quantities or values. These can influence statistical measures, particularly averages.

20. Business Recommendation

Based on what we found during the analysis, the retailer could use a **data-driven approach to inventory and sales management**.

The company should:

- Keep a close watch on high-revenue products.
- Use monthly sales patterns when planning inventory.
- Track returns and cancellations separately.
- Investigate unusually large transactions.
- Improve the collection of customer information.
- Analyze international markets separately to understand their potential.

Main Recommendation

The retailer should focus inventory planning on high-revenue products and periods with stronger sales, while keeping a separate track of returns, cancellations, and unusual transactions.

This recommendation connects our data-quality findings with the sales patterns we identified during the analysis.

21. Outputs

Online Retail.xlsx(application/vnd.openxmlformats-officedocument.spreadsheetml.sheet) -

23715344 bytes, last modified: 9/21/2026 - 100% done

Saving Online Retail.xlsx to Online Retail.xlsx

Dataset Shape: (541909, 8)

	Invoice No	StockCo de	Descripti on	Quanti ty	InvoiceD ate	UnitPri ce	Customer ID	Countr y
--	---------------	---------------	-----------------	--------------	-----------------	---------------	----------------	-------------

0	536365	85123A	WHITE HANGIN G HEART T-LIGHT HOLDER	6	2010-12-0 1 08:26:00	2.55	17850.0	United Kingdo m
1	536365	71053	WHITE METAL LANTER N	6	2010-12-0 1 08:26:00	3.39	17850.0	United Kingdo m
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-0 1 08:26:00	2.75	17850.0	United Kingdo m
3	536365	84029G	KNITTE D UNION FLAG HOT WATER BOTTLE	6	2010-12-0 1 08:26:00	3.39	17850.0	United Kingdo m
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-0 1 08:26:00	3.39	17850.0	United Kingdo m

DATA TYPES

	0
InvoiceNo	object
StockCode	object
Description	object
Quantity	int64

InvoiceDate	datetime64[ns]
UnitPrice	float64
CustomerID	float64
Country	object

dtype: object

MISSING VALUES

	0
InvoiceNo	0
StockCode	0
Description	1454
Quantity	0
InvoiceDate	0
UnitPrice	0
CustomerID	135080
Country	0

dtype: int64

DUPLICATE ROWS: 5268

STATISTICAL SUMMARY

	coun t	uni que	top	fre q	mean	min	25 %	50 %	75 %	max	std
Invoic eNo	5419 09.0	259 00.0	57358 5.0	111 4.0	NaN	NaN	Na N	NaN	Na N	NaN	NaN
Stock Code	5419 09	407 0	85123 A	231 3	NaN	NaN	Na N	NaN	Na N	NaN	NaN

Descri ption	5404 55	422 3	WHI TE HAN GING HEA RT T-LIG HT HOL DER	236 9	NaN	NaN	Na N	NaN	Na N	NaN	NaN
Quant ity	5419 09.0	NaN	NaN	NaN	9.55225	-809 95.0	1.0	3.0	10.0	809 95.0	218.08 1158
Invoic eDate	5419 09	NaN	NaN	NaN	2011-07-04 13:34:57.15 6386048	2010 -12-0 1 08:2 6:00	201 1-03 -28 11:3 4:00	201 1-07 -19 17:1 7:00	201 1-10 -19 11:2 7:00	201 1-12 -09 12:5 0:00	NaN
UnitPr ice	5419 09.0	NaN	NaN	NaN	4.611114	-110 62.0 6	1.25	2.08	4.13	389 70.0	96.759 853
Custo merID	4068 29.0	NaN	NaN	NaN	15287.6905 7	1234 6.0	139 53.0	151 52.0	167 91.0	182 87.0	1713.6 00303
Count ry	5419 09	38	Unite d Kingd om	495 478	NaN	NaN	Na N	NaN	Na N	NaN	NaN

	Missing Values	Percentage
InvoiceNo	0	0.00
StockCode	0	0.00
Description	1454	0.27
Quantity	0	0.00

InvoiceDate	0	0.00
UnitPrice	0	0.00
CustomerID	135080	24.93
Country	0	0.00

Negative quantity records: 10624

Invalid/zero price records: 2517

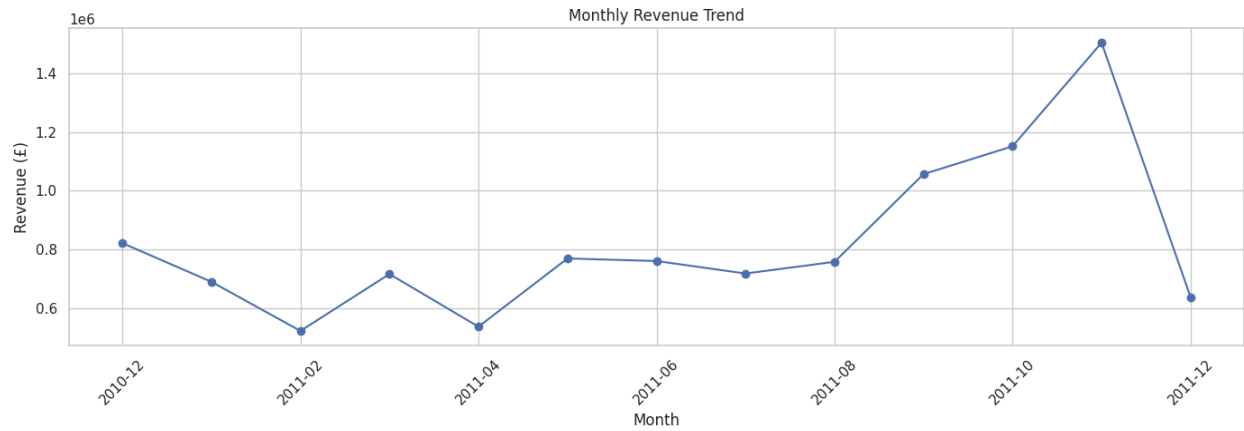
Cancellation invoices: 9288

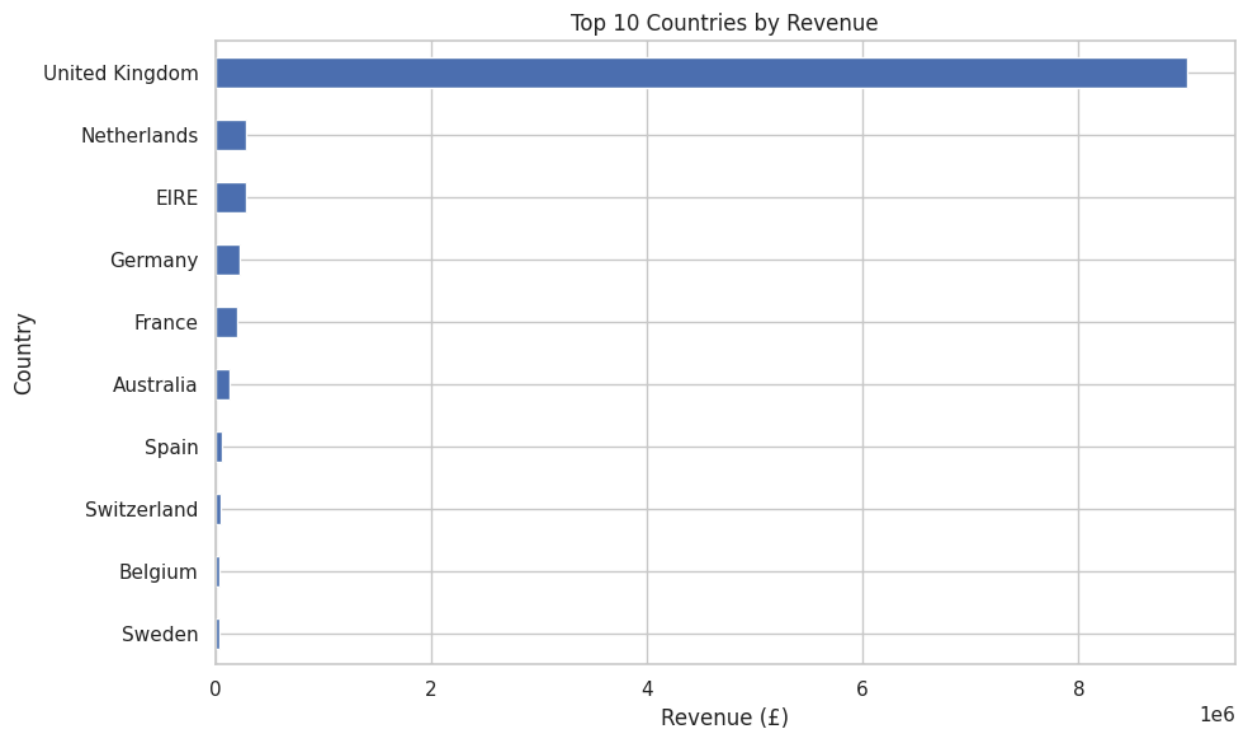
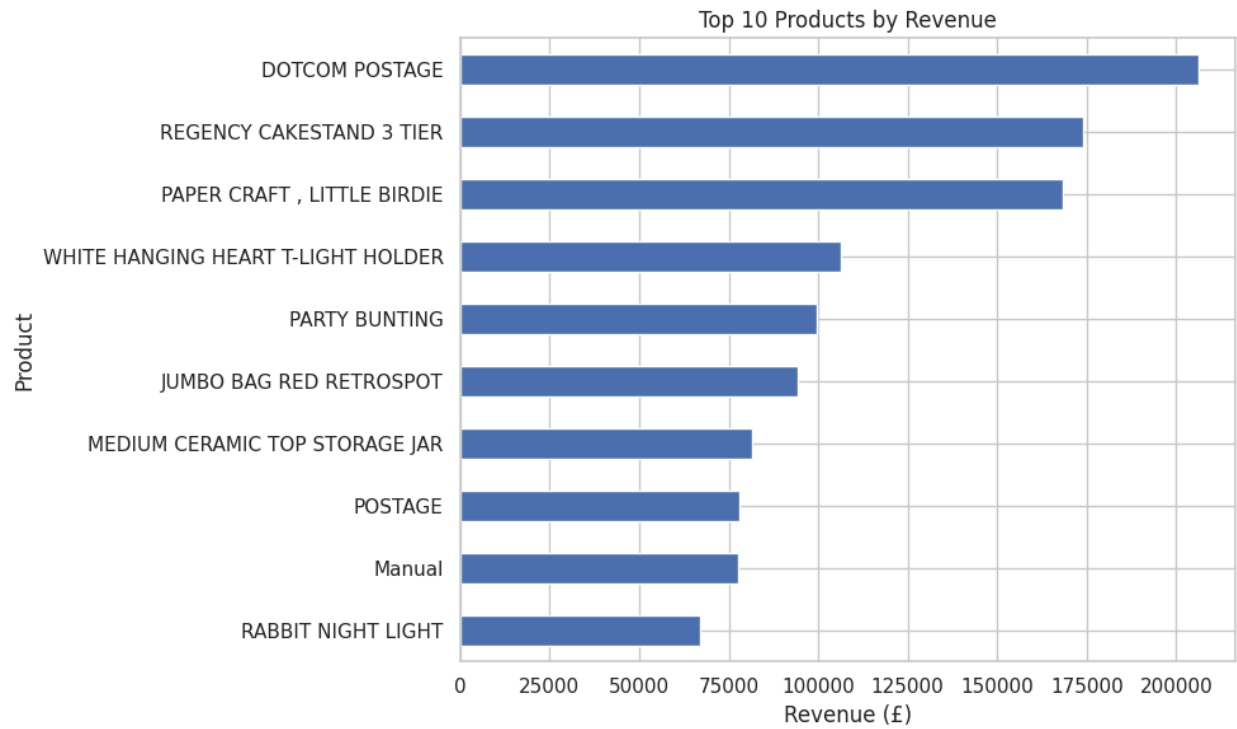
Duplicate rows: 5268

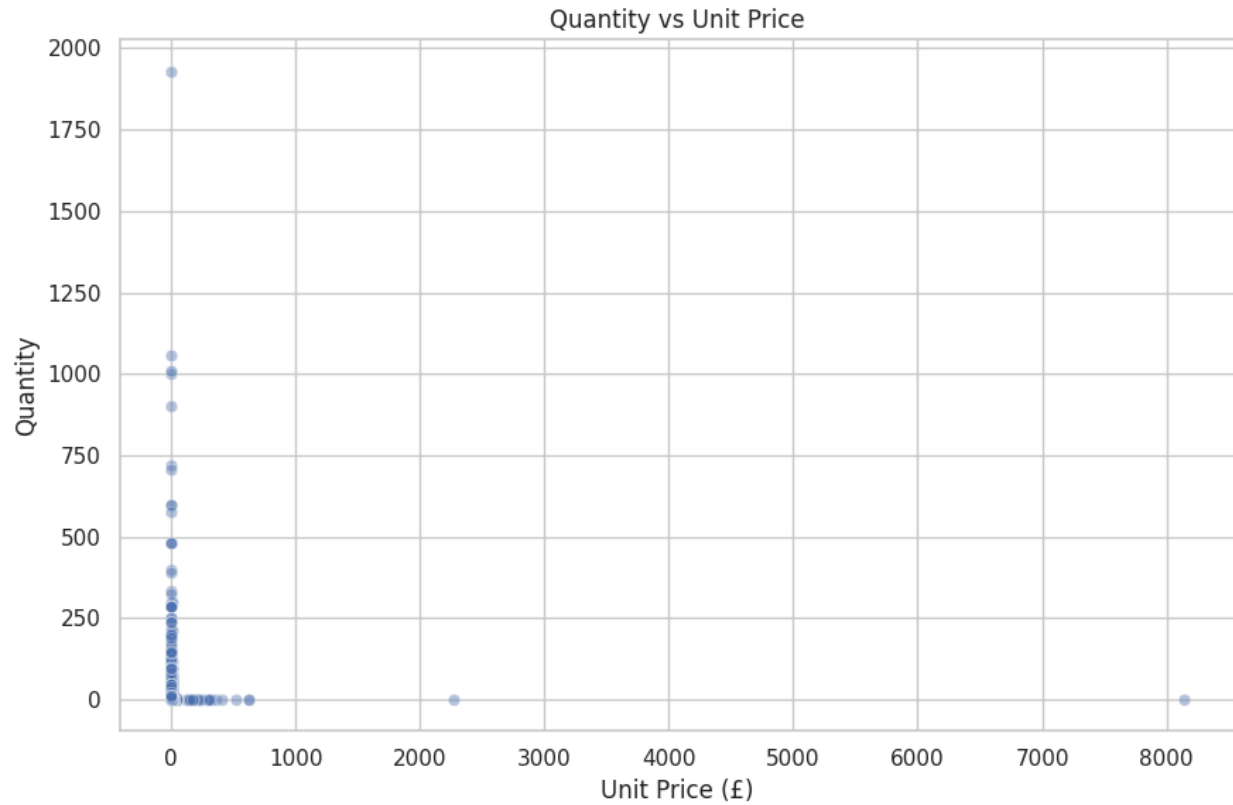
Clean dataset: (524878, 11)

	Invoi ceNo	Stock Code	Descri ption	Qua ntity	Invoice Date	Unit Price	Custo merID	Cou ntry	Reve nue	Mo nth	Day
0	536365	85123A	WHIT E HANG ING HEAR T T-LIG HT HOLD ER	6	2010-12-01 08:26:00	2.55	17850.0	Unite d King dom	15.30	2010-12	Wedn esday
1	536365	71053	WHIT E META L LANT ERN	6	2010-12-01 08:26:00	3.39	17850.0	Unite d King dom	20.34	2010-12	Wedn esday
2	536365	84406B	CREA M CUPI D HEAR TS	8	2010-12-01 08:26:00	2.75	17850.0	Unite d King dom	22.00	2010-12	Wedn esday

			COAT HANG ER								
3	53636 5	84029 G	KNITT ED UNIO N FLAG HOT WATE R BOTT LE	6	2010-1 2-01 08:26:0 0	3.39	17850.0	Unite d King dom	20.3 4	201 0-1 2	Wedn esday
4	53636 5	84029 E	RED WOOL LY HOTTI E WHIT E HEAR T.	6	2010-1 2-01 08:26:0 0	3.39	17850.0	Unite d King dom	20.3 4	201 0-1 2	Wedn esday







	Quantity
Description	
PAPER CRAFT , LITTLE BIRDIE	80995
MEDIUM CERAMIC TOP STORAGE JAR	78033
WORLD WAR 2 GLIDERS ASSTD DESIGNS	54951
JUMBO BAG RED RETROSPOT	48371
WHITE HANGING HEART T-LIGHT HOLDER	37872
POPCORN HOLDER	36749
PACK OF 72 RETROSPOT CAKE CASES	36396
ASSORTED COLOUR BIRD ORNAMENT	36362
RABBIT NIGHT LIGHT	30739

MINI PAINT SET VINTAGE	26633
-------------------------------	-------

dtype: int64

	Revenue
CustomerID	
14646.0	280206.02
18102.0	259657.30
17450.0	194390.79
16446.0	168472.50
14911.0	143711.17
12415.0	124914.53
14156.0	117210.08
17511.0	91062.38
16029.0	80850.84
12346.0	77183.60

dtype: float64

Average transaction value: £ 533.17

Median transaction value: £ 303.3

22. Conclusion

The **Data Detective Mission** gave us a practical experience of how raw business data can be explored and turned into useful information.

We started by understanding the dataset and checking its quality. We found issues such as missing customer information, duplicate records, cancelled transactions, and unusual values. After cleaning the data, we were able to analyze the sales information more reliably.

We then used statistics and visualizations to understand **monthly sales, product performance, country-wise revenue, and unusual transaction patterns**. These helped us see where the business was performing strongly and which areas needed closer attention.

Overall, this activity showed us that **data cleaning is just as important as data analysis**. Once the data was properly prepared, it became much easier to identify meaningful patterns and connect them to real business decisions, such as better inventory planning, monitoring returns, and focusing on high-performing products.